

Prathamesh Kulkarni

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EDUCATION

Ph.D. in Chemical and Biomolecular Engineering, GPA: 3.95/4.00

Rice University

Defended February 2026

Houston, TX

B.Tech. Chemical Engineering, CPI: 8.10/10.00

Indian Institute of Technology (IIT) Bombay

2019

Mumbai, India

SKILLS

Professional Summary: Trained chemical engineer with **5+ years of experience** in computational modeling and simulation of complex physical and process-relevant problems in thermodynamics, materials, transport, and reaction engineering

Programming: C++, Python, Bash, High-Performance Computing (SLURM) for remote parallel processing

Modeling & Simulation: Coarse-grained simulations, numerical methods, nonlinear optimization, energy minimization, gradient-based calculations/solvers, periodic lattice/off-lattice geometries with boundary conditions, computational mechanics/rheology simulations of soft materials (Discrete-Element/Finite-Element adjacent Methods)

Chemical Engineering & Dynamic Modeling: Monte Carlo techniques, Stochastic processes, Brownian dynamics, particle-based simulations like gas-liquid phase equilibria, Differential equations (ODE/PDE) for modeling for transport, chemical kinetics for reaction engineering modeling

Publishing: L^AT_EX, Microsoft Office, Inkscape, Data Visualization with Python (Matplotlib) and MATLAB

RESEARCH EXPERIENCE

PhD Candidate, Rice University

Houston Texas

Advisors: [Dr. Frederick MacKintosh](#) and [Dr. Anatoly Kolomeisky](#)

Spring 2020 - Present

- Developed C++ computational models of crosslinked fiber-based materials using lattice/off-lattice models to predict stress-strain response, rheology, and deformation-driven changes in soft material behavior
- Built a custom C++ stochastic simulation using Gillespie algorithm for force-dependent material fragmentation (event-driven fiber breakage, topology changes), quantifying rigidity thresholds, stress relaxation over time
- Demonstrated orders of magnitude increase in shear modulus of composites with “rope-like” filaments and shift in onset of rigidity/material stiffening to lower strains enabling requirement-specific material design
- Implemented nonlinear minimization algorithms (FIRE, gradient-based methods) in C++ to compute equilibrium network configurations under deformation, ensuring numerical stability and convergence
- Utilized HPC clusters (SLURM) to execute large-scale parallel simulations and MATLAB for post-processing

Undergraduate Research Assistant, Indian Institute of Technology Bombay

Mumbai, India

Advisor: [Dr. Supreet Saini](#), Department of Chemical Engineering

Dec 2017 – Feb 2018

- Developed a coupled ODE-based MATLAB framework for gene-regulatory response under sequential operating time windows of signals, and used constrained nonlinear optimization to identify parameter regimes where response crosstalk improves cost/benefit tradeoff (cellular fitness)

ACADEMIC PROJECTS

Robophysics Bootcamp

Atlanta, GA

[Goldman Lab](#), Georgia Institute of Technology

14-16 Jul 2025

- Assembled and Arduino-programmed [5-linker robot](#) with servo motors, quantified distance traveled-per-cycle scaling to test geometric-phase effects on locomotion - intersection of robotics and soft matter

Phase Equilibria using Gibbs Ensemble Monte Carlo

Houston, TX

Advisor: [Dr. Walter Chapman](#), Rice University

Fall 2019

- Applied thermodynamic modeling principles and Monte Carlo methods to predict gas-liquid equilibrium for a model fluid and investigate phase separation and density profiles near critical points: crucial for oil/gas, chemical, and pharmaceutical process design

Transient exposure of protein binding DNA sites in the nucleosome

Mumbai, India

Advisor: [Dr. Ranjeeth Padinhateeri](#), Biosciences and Bioengineering, Indian Institute of Technology Bombay *Spring 2018*

- Developed bead-spring Brownian-dynamics simulations for DNA-histone nucleosomes, implementing Langevin integration with interacting potentials to simulate spontaneous partial unwrapping - project for a graduate-level course that focused on computational biology modeling

Growth-related Modeling of the GAL System in yeasts -*S. cerevisiae* And *K. lactis* Mumbai, India

Advisors: [Dr. Sharad Bhartiya](#) and [Dr. K. V. Venkatesh](#), Department of Chemical Engineering Aug 2018 – Apr 2019

- Reproduced the existing growth-associated dynamic model for GAL gene regulatory network in *S. cerevisiae* growing on glucose and galactose using differential equations for associated reaction kinetics
- Explored and modeled the growth-burden tradeoff due to synthesis of GAL proteins in *S. cerevisiae* and *K. Lactis*

PUBLICATIONS

Kulkarni, P., Kolomeisky, A. B., MacKintosh, F. C., Stress relaxation in spring networks via force-dependent stochastic severing (<https://arxiv.org/abs/2606.01752>)

Kulkarni, P., Gupta, S., Kolomeisky, A. B., MacKintosh, F. C., Mechanics of composite networks with rope-like filament (In preparation) (Website: prathameshk867.github.io)

TALKS AND POSTER PRESENTATIONS (SELECTED)

Presented a poster on Stress relaxation in spring networks via force-dependent stochastic severing at the Smalley Curl Institute Colloquium 2024 (Best poster award)

Presented a talk on Mechanics of composite networks with rope-like filaments at the Global Summit (March Meeting) of the American Physical Society 2025: [talk abstract link](#)

LEADERSHIP AND OUTREACH

Outreach Chair — Rice University Houston Texas
ChBE Graduate Student Association (ChBE GSA) Jun 2021 – Jun 2022

- Acted as a liaison between ChBE GSA, AICHE, and ChBE Alumni committee and helped organize career fair preparation events such as Resume review, Mock Interviews, Fishbowl interview, and Career Panel for students

Treasurer — Rice University Houston Texas
ChBE Graduate Student Association (ChBE GSA) Jun 2020 – Jun 2021

- Managed the budget of ChBE GSA and applied for grants to support social events virtually for graduate student engagement and morale during the pandemic

Events Manager, Indian Institute of Technology Bombay Mumbai, India
AZeotropy - Chemical Engineering Symposium Apr 2017 – March 2018

- Coordinated a 2-tier team to conceptualize and execute 20+ events—lectures, panel discussions, exhibitions, and fireside chats—in partnership with industry and academic leaders, drawing over 2000 attendees
- Organized hands-on workshops on engineering simulation software (ASPEN, ANSYS, COMSOL, MATLAB) - creating educational opportunities for students to learn modeling outside of the curriculum

TEACHING EXPERIENCE

Teaching Assistant for Fluid Mechanics and Transport Phenomena CHBE 501 Fall 2021, 2022

Teaching Assistant for Computational Methods in Chemical Engineering Spring 2020

Teaching Assistant for Thermodynamics I, ChBE 411 Spring 2023

- Held office hours weekly to guide students and clarify concepts outside of class, and graded assignments for first-year 30+ graduate students and programming assignments for undergraduate students

INDUSTRIAL TRAINING

Modeling and Simulation of Catalytic Converter Pune, India
Instructor: [Harsh Vinjamoor](#), Manager R&D SEDEMAC Mechatronics May 2017 – Jul 2017

- Analyzed the chemical reaction kinetics involved in catalytic converter and simulated surface reactions via Python
- Studied and attempted to replicate outlet gas concentration profiles variations with changes to air-to-fuel ratio at the inlet of catalytic converter - valuable understanding to design emission control strategies

AWARDS & ACHIEVEMENTS

Recipient of Graduate Student Travel Grant award by Rice Engineering Alumni (REA) Spring 2025

Recipient of Platinum Poster Award at the Smalley Curl Institute Colloquium Summer 2024

Honored with Undergraduate Research Award for exemplary contribution to research 2018

Awarded Department Color for exceptional contribution to the chemical engineering symposium activities 2018

RELEVANT COURSEWORK

Operations Management, Advanced Numerical Methods, Modeling Biological Systems/Processes, Statistical Mechanics, Physicochemical Hydrodynamics, Artificial Intelligence (AI) in Process Engineering, Fluids & Transport Phenomena